The Economics of Generative AI

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Abstract

Generative AI, advanced from traditional AI that focuses on specific work, has fundamentally altered the economics of AI and data. We've developed a dynamic model to illustrate how two key factors—non-rival data and rival computing power—shape the economics of generative AI and reveal market inefficiencies. The economy faces risks such as excessive investment in computing, limitations in generative AI development, and distorted data sharing.

Keywords: generative AI, economics of AI, computing power, storage, data

1 Introduction

The advent of Artificial Intelligence (AI) has significantly reshaped the economic landscape, prompting economists to develop a novel framework for analyzing emerging phenomena, potential risks, and regulatory needs in the AI-driven economy. Existing literature has predominantly focused on how AI enhances productivity, fosters innovation, influences the labor market, and drives growth (Aghion et al., 2017; Agrawal et al., 2019). However, much of this discourse has centered on traditional AI, also referred to as weak AI, which notably differs from the emergence of generative AI, prominently seen in 2022. The ascendancy of generative AI has broadened AI capabilities from merely executing intelligent tasks based on predefined parameters to creating original content through interaction with humans, necessitating the seamless integration of data input, storage, and computational power. Table 1 presents a succinct comparison between traditional AI and generative AI.

The key element of traditional AI is the algorithm, whereas generative AI relies more on the combination of data and algorithm. While a well-trained traditional AI requires significant computing power and extensive datasets during development, its operational usage does not

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Characteristics	Traditional AI	Generative AI
Key Elements	Algorithm	Data + Algorithm
Main Applications	Specific tasks	Innovation and creation
High-Quality Input Requirements	Minimal	Substantial
Data, Storage, and Computing Requirements	Only for development	Development + Application

Table 1: Comparison between traditional AI and generative AI

necessarily demand vast amounts of data input, computing power, or data storage. A model driven by traditional AI can be easily deployed on a personal computer or portable device, and applied in various scenarios including identity recognition, fraud detection, and forecasting. In contrast, generative AI finds predominant use in textual processing, graphic generation, and product design. Its practical application often entails interactive training with users and continuous input of high-quality data even after the model's release.

Recent news and industry reports highlight significant privacy concerns and unexpected misallocation of computing power due to generative AI. [1] Unlike traditional AI, which typically produces outputs with clear boundaries, generative AI faces the risk of unauthorized use of users' confidential inputs to generate content, potentially leading to the dissemination of copyrighted material. Additionally, both the training and content generation processes of generative AI require substantial computing power, resulting in a rapid increase in computing costs and hindering the development of other fields such as pharmaceuticals and academic research. This situation raises concerns about the misallocation of computing power from a societal perspective.

Based on these characteristics, we develop a dynamic model that reveals the role of data input, computing power, and data storage in the economy of generative AI: consumers contribute data to data intermediaries to develop data products, consume final products, and encounter privacy concerns. Data intermediaries engage in noncompetitive sales of data products to support the development and application of generative AI, employing third-degree price discrimination, and bearing the costs of data storage. Both the development and application of generative AI consume rivalrous computing power while providing final goods producers with generated content.

Our results have unveiled several market inefficiencies and elucidated the interplay between

[1] For instance, the privacy leakage of ChatGPT in April 2024, as reported by The Paper: https://www.thepaper.cn/newsDetail˙forward˙22644353.

market distortions, monopolistic markups, inadequate development of generative AI, and the depletion of computing power. Through our comparative statics analysis, we observe that economies endowed with greater knowledge and higher output elasticity of data tend to exhibit more pronounced market inefficiencies. This finding resonates with Marston et al. (2011), highlighting the suboptimal allocation of data storage and computing power. Furthermore, we delve into the issues of data scarcity and misuse across diverse economies within a unified framework, departing from the predominant focus on privacy concerns or data externalities, as discussed by Jones and Tonetti (2020) and Cong et al. (2021). Our comparative statics analysis also explores the impact of technological advancements, such as encryption technology and chip development, on the economy of generative AI.

Except for prospective research on the economics of AI, researches on data economy is quite related to our paper, encompassing the economic valuation, attributes, and privacy implications of data factor. Jones and Tonetti (2020) pioneerly introduce the concept of data nonrivalry in production, dissecting the impacts of distinct data properties on economic growth. Cong et al. (2021, 2022) focalize on the "data desensitizing" process within the innovation, showing how privacy concerns are mitigated as raw data transforms into knowledge. The former scrutinizes the potential scarce of labor in the innovation sector and data overuses, while the latter investigates deficient data sharing arising from monopolistic patent rights and low data returns. Begenau et al. (2018) and Farboodi and Veldkamp (2021) scrutinize the function of data, including its role in production estimation and alleviating information asymmetry. Distinct from prior literature, our present study explores data-sharing distortions stemming from the interaction of data, computing power, and data storage. Furthermore, our insights into data externalities surpass discussions in Ichihashi (2021) and Acemoglu et al. (2022). Our paper also has close connection with literature on data production (Huang et al., 2023), data infrastructure (Hou et al., 2022), data privacy (Goldfarb and Tucker, 2011; Abowd and Schmutte, 2019; Fainmesser et al., 2023), cloud service pricing (Dierks and Seuken, 2022), and data regulation (He et al., 2022).

2 Model Setup

To begin, we establish a two-period static framework encompassing five distinct sectors: consumers, final good producers, developers and producers of generative AI computing power producers, data storage producers, and data intermediaries.

2.1 Consumers

We assume the presence of a continuum of uniform consumers spanning the interval $[0, 1]$. Each consumer contributes an indivisible unit of labor and receives compensation denoted as wage w. Simultaneously, they sell data d^s in the market at the price p_d and receive profits as shareholders. The superscript s of d^s indicates supply. For normalization, the price of the final good is set at 1. The optimization problem faced by consumers is expressed as follows:

$$\max_{c,d^s} \ln c - \phi(d^s) \tag{1}$$

where c is the consumption of final good, and $\phi(d^s)$ encapsulates the privacy cost linked to data sales. The budgetary constraint is articulated as:

$$c \leq w + \sum_{\nu=1}^{N} \pi_\nu + p_d d^s \tag{2}$$

The first order condition with respect to d^s takes the form:

$$\frac{p_d}{c} = \phi'(d^s) \tag{3}$$

2.2 Producers

The producer segment encompasses a representative final goods producer and N generative AIs. The production function for the final goods producer exhibits a constant elasticity of substitution:

$$Y = (\sum_{\nu=1}^{N} Y_\nu^{\frac{\gamma-1}{\gamma}})^{\frac{\gamma}{\gamma-1}} \tag{4}$$

The magnitude of the final goods output Y scales with the amount of generative AIs N and is contingent upon the parameter γ. The optimization objective of the final goods producer is delineated as follows:

$$\max_{Y_\nu} \left\{ \left[\sum_{\nu=1}^{N} (Y_\nu)^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}} - \sum_{\nu=1}^{N} p_\nu Y_\nu \right\} \tag{5}$$

where p_ν denotes the price of each generative AI.

The pursuit of profit maximization yields the relationship:

$$p_\nu = (\frac{Y}{Y_\nu})^{\frac{1}{\gamma}}. \tag{6}$$

Moving on to the usage of generative AIs, the formulation stands as:

$$Y_\nu = D_{1,\nu}^{\eta_1} K_{c1,\nu}^{1-\eta_1} \tag{7}$$

Within this context, $D_{1,\nu}$ denotes data procured from data intermediaries, $K_{c1,\nu}$ denotes computing power, and $\eta \in (0,1)$ captures the significance of data in the application of generative AIs. The optimization problem for profit maximization takes the following shape:

$$\pi_\nu = \max_{D_{1,\nu}, K_{c1,\nu}} p_\nu Y_\nu - p_{D1} D_{1,\nu} - p_{K_c} K_{c1,\nu} \tag{8}$$

where p_{K_c} denotes the computing power price.

The first-order conditions pertaining to $D_{1,\nu}$ and $K_{c1,\nu}$ can be rearranged as follows:

$$(1 - \frac{1}{\gamma})\eta_1 p_\nu Y_\nu = p_{D1} D_{1,\nu} \tag{9}$$

$$(1 - \frac{1}{\gamma})(1 - \eta_1) p_\nu Y_\nu = p_{K_c} K_{c1,\nu} \tag{10}$$

Moreover, we have

$$\pi_\nu = \frac{1}{\gamma}(\frac{Y}{Y_\nu})^{\frac{1}{\gamma}} Y_\nu \tag{11}$$

2.3 R&D of Generative AI

In this context, we posit the existence of a substantial number of potential market developers of generative AI. These developers of generative AI engage in the procurement of data D_2 and computing power K_{c2} to foster novel generative AIs and thereby accrue monopolistic profits. The production function characterizing the development of generative AI is formulated as:

$$\Delta N = N_0^\theta D_2^{\eta_2} K_{c2}^{1-\eta_2} \tag{12}$$

where $\theta \in [0,1]$ captures knowledge spillover effects, while η_2 situated within the interval $(0,1)$ signifies the relevance of data within the the development of generative AI. The initial amount of generative AI is set at N_0.

After successful development of generative AIs, these models are subsequently employed in various specific applications (e.g., translation, text refinement, code debugging). Both the development and application of generative AIs consume data and computing power resources, thereby necessitating a focus on the optimal reassignment of these resources.

Given the profit associated with a successful development π, the optimization problem faced by an developer takes the form:

$$\max_{D_2, K_{c2}} \Delta N\pi - p_{D2}D_2 - p_{K_c}K_{c2} \tag{13}$$

The first-order conditions concerning D_2 and K_{c2} can be expressed as:

$$\eta_2 \Delta N\pi = p_{D2}D_2 \tag{14}$$

$$(1-\eta_2)\Delta N\pi = p_{K_c}K_{c2,t} \tag{15}$$

In the equilibrium, $\pi = \pi_\nu$ and $N = N_0 + \Delta N$.

2.4 Production of computing power

Assuming the production of unit computing power requires labor equivalent to $l_c = \chi_c K_c$. Given the price of computing power p_{K_c}, the profit maximization problem faced by a computing power producer can be articulated as:

$$\max_{K_c^s} p_{K_c}K_c^s - w\chi_c K_c^s \tag{16}$$

2.5 Data Intermediaries and Data Storage

Drawing from the insights of Jones and Tonetti (2020) and Cong et al. (2021), we introduce a data intermediary into the framework. This intermediary buys data d from consumers at a price p_d, employs labor $l_s = \chi_s K_s$ with a wage w, and generates data storage K_s. Subsequently, the intermediary sells aggregated data to both users and potential developers of generative AIs. In light of data's nonrival nature, the intermediary can engage in third-degree price discrimination when selling data. Specifically, users of generative AIs acquire data at a price p_{D1}, whereas potential developers of generative AI obtain data at a price p_{D2}.[2] Under conditions of open market entry, the data intermediary's profits are nullified. Consequently, we leverage a zero-profit condition alongside a cost-minimization problem to encapsulate the intermediary's optimization task:

$$\min_{K_s, d} w\chi_s K_s + p_d d \tag{17}$$

[2]Implicitly, this assumes no data resale occurs among users and developers of generative AI.

subject to the constraint

$$D_i^s \le \min\{d, K_s\}, \quad i \in \{1, 2\} \tag{18}$$

$$p_{D1} \sum_{\nu=1}^{N_0+\Delta N} D_{1,\nu}^s + p_{D2} D_2^s = w\chi_s K_s + p_d d \tag{19}$$

Figure 1 provides an illustrative overview of our model's configuration.

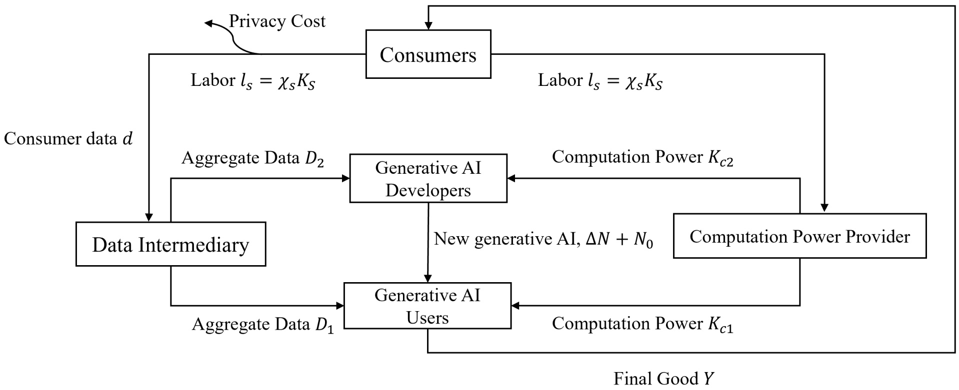


Figure 1: Model Setup

2.6 Equilibrium

The decentralized equilibrium solution encompasses the variables $\{c, d^s, d, Y, Y_\nu, N, D_{1,\nu}, D_{1,\nu}^s, D_2^s, D_2, K_s, K_{c1,\nu} K_{c1,\nu}^s, K_{c2}, K_{c2}^s\}$ as well as the wage rates $w, p_d, p_D, p_{K_c}, p_\nu$. In this equilibrium, $\{c, d^s\}$ optimize consumer utility, $\{Y_\nu, D_{1,\nu}, K_{c1,\nu}\}$ maximize the profit of final goods producers, $\{N, D_2, K_{c2}\}$ optimize the development of generative AIs, $\{K_{c1,\nu}^s, K_{c2}^s\}$ solve the computing power producer problem, and $\{D_{1,\nu}^s, D_2^s, K_s\}$ optimize the data intermediaries' tasks.

The market equilibrium conditions encompass:

1. Final goods market: $Y = c$.
2. Labor market: $1 = \chi_s K_s + \chi_c K_c^s$.
3. Computing power market: $K_c^s = K_{c2} + \sum_{\nu=1}^{N} K_{c1,\nu}$.
4. Data market between data intermediaries and consumers: $d^s = d$.
5. Data market between data intermediaries and other entities: $D_{1,\nu} = D_{1,\nu}^s, D_2 = D_2^s$.

These equilibrium conditions collectively define the intricate interplay of various factors, ensuring the equilibrium state of the economics of generative AI.

3 Theoretical Analysis

3.1 Social Optimal Allocation

Let $K_{c1} \equiv NK_{c1,\nu}$, denote the production used in production. The optimization problem for a social planner can be formulated as follows:

$$\begin{aligned} & \max \ln c - \phi(d) \\ s.t. \quad & c = Y = (N_0 + \Delta N)^{\frac{\gamma}{\gamma-1} - (1-\eta_1)} D^{\eta_1} K_{c1}^{1-\eta_1} \\ & \chi_s K_s + \chi_c K_c = 1 \\ & \Delta N = N^\theta D^{\eta_2} K_{c2}^{1-\eta_2} \\ & K_{c1} + K_{c2} = K_c \\ & D = \min\{d, K_s\} \end{aligned} \tag{20}$$

Here, the exponent of the number of AIs $(N_0) + \Delta N$ comprises two components: (i) $\frac{\gamma}{\gamma-1} > 1$ derives from the welfare enhancement resulting from the development of generative AI and the expansion of product variety; (ii) $-(1-\eta_1)$ is a consequence of the rivalry inherent in computing power. For instance, when a substantial language model employs processing power to compute data, the same processor cannot be used by any other process, which contrasts the non-rivalry characteristic of data usage. With increased development of generative AI, computing power becomes distributed among more generative AIs, leading to the dilution of computing power's productivity by a factor of $N^{-(1-\eta_1)}$. This implies that in the context of AI development, the principle of "more is better" doesn't apply to the number of generative AIs, as the efficiency of each model could significantly deteriorate due to computing power scarcity. However, the economy can optimally harness data without diminishing its value due to data's non-rivalrous nature.

Consequently, we define $\Gamma = \frac{\gamma}{\gamma-1} - (1-\eta_1) > 0$ to characterize the welfare improvement stemming from the development of generative AIs.

Consider the variable $\kappa_{c2} \equiv K_{c2}/K_c$. The optimization problem in Equation (20) can be equivalently expressed as:

$$\max \Gamma \ln(N_0 + N^\theta K_s^{\eta_2} (\kappa_{c2} K_c)^{1-\eta_2}) + \ln(K_s^{\eta_1} \left[(1-\kappa_{c2}) K_c\right]^{1-\eta_1}) - \phi(K_s) \tag{21}$$

subject to the constraint $\chi_s K_s + \chi_c K_c = 1$. We establish the following proposition:

Proposition 1. *In the social optimal allocation, $l_s^* = \chi_s K_s^*$ and κ_{c2}^* are given by the following* *system:*

$$\begin{cases} \dfrac{\Gamma\eta_2(N^*-N_0)}{N^* l_s^*} + \dfrac{\eta_1}{l_s^*} = \dfrac{\phi'(d^*)}{\chi_s} + \dfrac{\Gamma(1-\eta_2)(N^*-N_0)}{N^*(1-l_s^*)} + \dfrac{1-\eta_1}{1-l_s^*} \\ \dfrac{\Gamma(1-\eta_2)(N^*-N_0)}{N^*\kappa_{c2}^*} = \dfrac{1-\eta_1}{1-\kappa_{c2}^*} \end{cases} \tag{22}$$

where $N^* = N_0 + N_0^\theta (\frac{l_s^*}{\chi_s})^{\eta_2} (\kappa_{c2}^* K_c^*)^{1-\eta_2}$, $d^* = D^* = K_s^* = \frac{l_s^*}{\chi_s}$, $K_c^* = \frac{1-l_s^*}{\chi_c}$, *with other variables implicitly expressed in terms of* (l_s^*, K_{c2}^*).

In Proposition 1, the first equation captures the labor allocation trade-off between the storage production sector and the computing power sector. The left-hand side represents the enhanced development of generative AI through increased allocation of labor to the data storage sector, coupled with augmented data storage. The right-hand side quantifies the opportunity cost of data storage, encompassing the marginal utility improvement from channeling more labor to the computing power sector and the associated privacy cost linked to data storage $\frac{\phi'(d^*)}{\chi_s}$. Similarly, the second equation elucidates the trade-off in computing power allocation between the development and application of generative AI. The left-hand side denotes the marginal utility improvement resulting from directing computing power to the the development of generative AI, while the right-hand side signifies the increased marginal utility from employing computing power in the application of generative AI. Owing to the rivalry intrinsic to computing power, the return of computing power in usage of generative AI represents the opportunity cost of the development of generative AI.

3.2 Decentralized Market Equilibrium

Considering the optimization pursued by market entities and the market clearing conditions, we deduce the ensuing proposition:

Proposition 2. *It is established that* $l_s^{**} = \chi_s K_s^{**}$, *and* κ_{c2}^{**} *is governed by the following system of equations:*

$$\begin{cases} \dfrac{\eta_2(N^{**}-N_0)}{\gamma N^{**} l_s^{**}} + (1-\dfrac{1}{\gamma})\dfrac{\eta_1}{l_s^{**}} = \dfrac{\phi'(K_s^{**})}{\chi_s} + \dfrac{(1-\eta_2)(N^{**}-N_0)}{\gamma N^{**}(1-l_s^{**})} + (1-\dfrac{1}{\gamma})\dfrac{1-\eta_1}{1-l_s^{**}} \\ \dfrac{(1-\eta_2)(N^{**}-N_0)}{\gamma N^{**}\kappa_{c2}^{**}} = (1-\dfrac{1}{\gamma})\dfrac{1-\eta_1}{1-\kappa_{c2}^{**}} \end{cases} \tag{23}$$

where $N^{**} = N_0 + N_0^\theta (\frac{l_s^{**}}{\chi_s})^{\eta_2} (\kappa_{c2}^{**} K_c^{**})^{1-\eta_2}$, $d^{**} = D^{**} = K_s^{**} = \frac{l_s^{**}}{\chi_s}$, $K_c^{**} = \frac{1-l_s^{**}}{\chi_c}$. $c^{**} = Y^{**} = (N^{**})^\Gamma (D^{**})^{\eta_1} [(1-K_{c2}^{**} K_c^{**}]^{1-\eta_1}$.

In the comparison between the equations in Proposition 2 and Proposition 1, it is evident that market distortions arise due to the monopolistic right aimed at promoting the development of generative AI. These distortions significantly impact the allocation of computing power κ_{c2}, data D, and aggregated computing power K_c. In particular, within a competitive market equilibrium, the marginal return of AI-generated content is $1 - \frac{1}{\gamma}$, while the marginal return of the development of generative AI is $\frac{1}{\gamma}$. In contrast, within the social planner's equilibrium, the respective marginal returns stand at 1 and $\Gamma = \frac{\gamma}{\gamma-1} - (1 - \eta_1)$. This discrepancy underlines three critical observations:

Distorted development of generative AI and inefficiency: in the context of market competition, potential developers of generative AI do not account for the dilution of computing power. Notably, the marginal return of the development of generative AI as derived in Proposition 2 lacks the inclusion of $-1 + \eta_1$, in contrast to its consideration in the socially optimal scenario. Consequently, an overabundance of developers of generative AI could lead to inefficiencies within the economy.

Allocation distortions due to monopolistic profits: the allocation of computing power and labor within a decentralized equilibrium stems from the marginal return of incumbents and potential developers of generative AI, diverging from the marginal utility of consumers in a socially optimal allocation. As illuminated by Proposition 2, the incumbent's markup manifests as $\frac{1}{\gamma}$, thereby yielding a return of computing power and data in AI application at $1 - \frac{1}{\gamma}$. Conversely, from the AI developers' standpoint, the return of these factors is contingent upon the development of generative AI success, being determined by $\frac{1}{\gamma}$. In essence, monopolistic profits attained through the development of generative AI exert a distorting influence on resource allocation.

Reformulating the allocation of computing power as in Proposition 2 results in the equation:

$$\frac{(1 - \eta_2)N^{**}}{(\gamma - 1)N^{**}\kappa_{c2}^{**}} = \frac{1 - \eta_1}{1 - \kappa_{c2}^{**}} \tag{24}$$

Considering a given data quantity D, a comparative analysis between this and the social planner's computing power allocation brings forth two scenarios:

1. Inefficiency Dominance ($\Gamma < \frac{1}{\gamma-1}$): in instances where Γ falls below the threshold of $\frac{1}{\gamma-1}$, the inefficiency highlighted in the first implication prevails. The decentralized equilibrium results in an excessive allocation of computing power towards the development of generative AI, thereby diluting the available computing power pool due to an abundance of developers of generative AI.

2. Monopolistic Profit Dominance ($\Gamma > \frac{1}{\gamma-1}$): conversely, when Γ surpasses $\frac{1}{\gamma-1}$, the second

inefficiency holds sway. In such cases, the computing power directed towards the development of generative AI within the decentralized equilibrium proves insufficient. The meager monopolistic profits fail to entice developers of generative AI to secure an adequate computing power allocation.

In essence, the monopolistic profit incentives associated with the development of generative AI can lead to either excessive or inadequate computing power allocation, depending on the relative significance of Γ and $\frac{1}{\gamma-1}$.

Simultaneous impact on labor and privacy: a further insight stems from the labor allocation equation as demonstrated in Proposition 2. Formally, equation (25) encapsulates two distinctive distortions, shedding light on both the development of generative AI and privacy costs:

$$\begin{aligned} &\underbrace{\frac{1}{\gamma-1}}_{\text{Distortion of AI R\&D}} \frac{1}{N^{**}} \frac{\eta_2(N^{**}-N_0)}{L_s^{**}} + \frac{\eta_1}{L_s^{**}} \\ =& \underbrace{\frac{\gamma}{\gamma-1}}_{\text{Distortion of Privacy Cost}} \frac{\phi'(K_s^{**})}{\chi_s} + \underbrace{\frac{1}{\gamma-1}}_{\text{Distortion of AI R\&D}} \frac{1}{N^{**}} \frac{(1-\eta_2)(N^{**}-N_0)}{1-L_s^{**}} + \frac{1-\eta_1}{1-L_s^{**}} \end{aligned} \tag{25}$$

Crucially, the key takeaway is that this distortion doesn't solely impact the marginal return of storage and computing power; rather, it concurrently influences the allocation of labor, aggregated computing power, and data volume. This interplay further underscores the intricate ramifications of monopolistic profits on resource distribution.

Furthermore, the price attributed to data doesn't adequately encompass the privacy cost $\frac{\phi'(d)}{\chi_s}$ due to the monopolistic markup $\frac{1}{\gamma}$. Consequently, consumers exhibit a propensity to curtail their data sharing. If the data volume remains consistent between the decentralized equilibrium and the optimal allocation, the decentralized equilibrium necessitates an augmented marginal cost for consumer privacy, amplifying it by a factor of $\frac{\gamma}{\gamma-1} > 1$.

This insight highlights the multifaceted dynamics through which monopolistic profit-driven distortions intertwine with both labor allocation and the valuation of privacy costs, amplifying the complexity of factors at play within the economy of generative AI.

3.3 Numerical Calibration

In this subsection, we delve into a comparative static analysis of parameters related to computing power and labor, employing numerical calibration. Notably, we assume a quadratic form for the privacy cost, i.e., $0.5\phi d^2$. For clarity, Panel A of Table 2 provides an overview of the pertinent parameters involved.

Table 2: Baseline Parameters and Numerical Example

Panel A: The Parameters in the Numerical Calibration		
Parameters	Definitions	Values
η_1	Weight assigned to data in application of generative AI	0.4
η_2	Weight assigned to data in the development of generative AI	0.4
χ_s	Coefficient governing relationship of labor to cost	0.5
χ_c	Production cost of computing power	0.5
γ	Elasticity parameter characterizing production	0.5
N_0	The initial amount of generative AIs	2
θ	Degree of knowledge spillover in the development of generative AI	0.85
ϕ	Coefficient quantifying privacy cost of data	0.2

Panel B: Difference between Social Optimum and Decentralized Equilibrium		
Variables	Social Optimal	Decentralized
Data d	0.6920	0.6523
Aggregated computing power K_c	1.3080	1.3476
Fraction of computing power used in generative AI R&D κ_{c2}	0.1375	0.0347
Increment of generative AIs $N^{**} - N_0$	0.5558	0.2420
Welfare from consumption $\ln c$	0.6132	0.5790
Privacy cost $-\phi(d)$	-0.0479	-0.0426
Total welfare $\ln c - \phi(d)$	0.5653	0.5365

Panel B of Table 2 offers a comprehensive summary of the calibration results, highlighting the contrasting outcomes between the social optimal allocation and the decentralized equilibrium. Notably, in the decentralized equilibrium, the amount of data allocated is slightly lower than that in the social optimal allocation. This phenomenon signifies a trend of labor allocation towards the production of processors, accompanied by an excessive investment in computing power. In contrast, the social optimal allocation dedicates a larger proportion of computing power to the the development of generative AI sector, a four-fold increase compared to the decentralized equilibrium. Although the aggregated computing power is higher in the decentralized equilibrium, the insufficiency of data leads to a deficiency in the development of generative AI, ultimately resulting in a welfare gap between the two scenarios. These calibration results underscore the importance of striking a balance between labor allocation, computing power distribution, and data utilization in order to achieve optimal outcomes within the digital economy.

Furthermore, we engage in a comparative static analysis to evaluate the parameter effects on the equilibrium. Initially, we focus on the impact of varying the initial development of generative AI (or initial knowledge), denoted as N_0, on the equilibrium, as depicted in the first two rows of Figure 2. Remarkably, with an increase in knowledge, both consumers within the social planner's paradigm and the decentralized economy exhibit a propensity to curtail their data sharing. This observation aligns with the findings of Cong et al. (2021), wherein heightened knowledge levels lead to diminished data demand, owing to the impetus provided for novel the development of generative AI and the concurrent mitigation of the overall privacy risk. A concomitant consequence of reduced data demand is an augmented allocation of labor towards computing power production, which, in turn, culminates in the enhanced exploitation of data, thereby catalyzing production and the development of generative AI. It is important to underscore, however, that due to limitations in data sharing and the inadequacy of computing power allocation within the the development of generative AI sector, a widening discrepancy in knowledge accumulation between the social optimal state and the decentralized equilibrium ensues as the initial knowledge stock escalates.

In the last two rows of Figure 2, we explore the impact of γ on the equilibrium outcomes. This sheds light on the complex relationship between market efficiency and certain key parameters.

Referring back to Proposition 1, when γ increases, it has a mixed impact on the social optimal equilibrium. On one hand, the increase weakens the positive influence of the development of generative AI on welfare, leading to a reduced focus on the development of generative AI-related sectors and subsequently a decreased demand for data in the economy. This aligns with the intuition that when the development of generative AI's role becomes less pronounced, the need for data to support that the development of generative AI decreases.

However, the decentralized equilibrium reacts in a more nuanced manner. As γ grows, the monopolistic markup $1/\gamma$ decreases, which curbs the motivation for the development of generative AI. This can lead to a reduction in the development of generative AI efforts. At the same time, as seen in Equation (25), this reduction in the monopolistic markup helps alleviate market distortions. This, in turn, allows for a higher compensation for consumer privacy, addressing concerns related to data privacy.

In the decentralized equilibrium, the interplay between these factors results in a more complex outcome. As γ increases, the amount of data D approaches a level closer to the optimal case. This suggests that while higher γ initially hampers the development of generative AI, the reduction in market distortions due to lower monopolistic markup has a balancing effect, nudging the data economy closer to an optimal state.

This finding highlights the intricate trade-offs and interactions between the development of generative AI incentives, market competition, and data usage, providing a fresh perspective on how changes in γ impact equilibrium outcomes in the data-driven economy.

Figure 3 sheds light on how changing the output elasticity of data affects the equilibrium, η_1 and η_2, which captures different AI algorithms are adopted. These figures help us see how different conditions impact the balance of the economy.

When we make data more important in production, in the first two rows of Figure 3, it increases the demand for data in production. This shift leads to more people working with data instead of just focusing on computing power. What's interesting is that due to data's unique nature, more nonrival data also encourages the development of generative AI to use it more. Because the demand for computing power in production declines, more computing power is directed toward the development of generative AI. This fosters the development of generative AI in the economy.

When η_1 is low, the economy doesn't use enough data; but when η_1 is high, it overuses data. This finding contrasts with previous theories. Normally, data scarcity is linked to how companies have too much control or how new ideas destroy old ones. Here, it's different. The balance between using computing power and data availability plays a role in this.

Furthermore, in the social optimal scenario, we notice that when η_1 goes up, so does the the development of generative AI benefit Γ. This means that if data can do more work in the development of generative AIs, it's good for everyone. But in reality, the decentralized scenario doesn't use this idea well. There's a shortage of computing power used in the development of

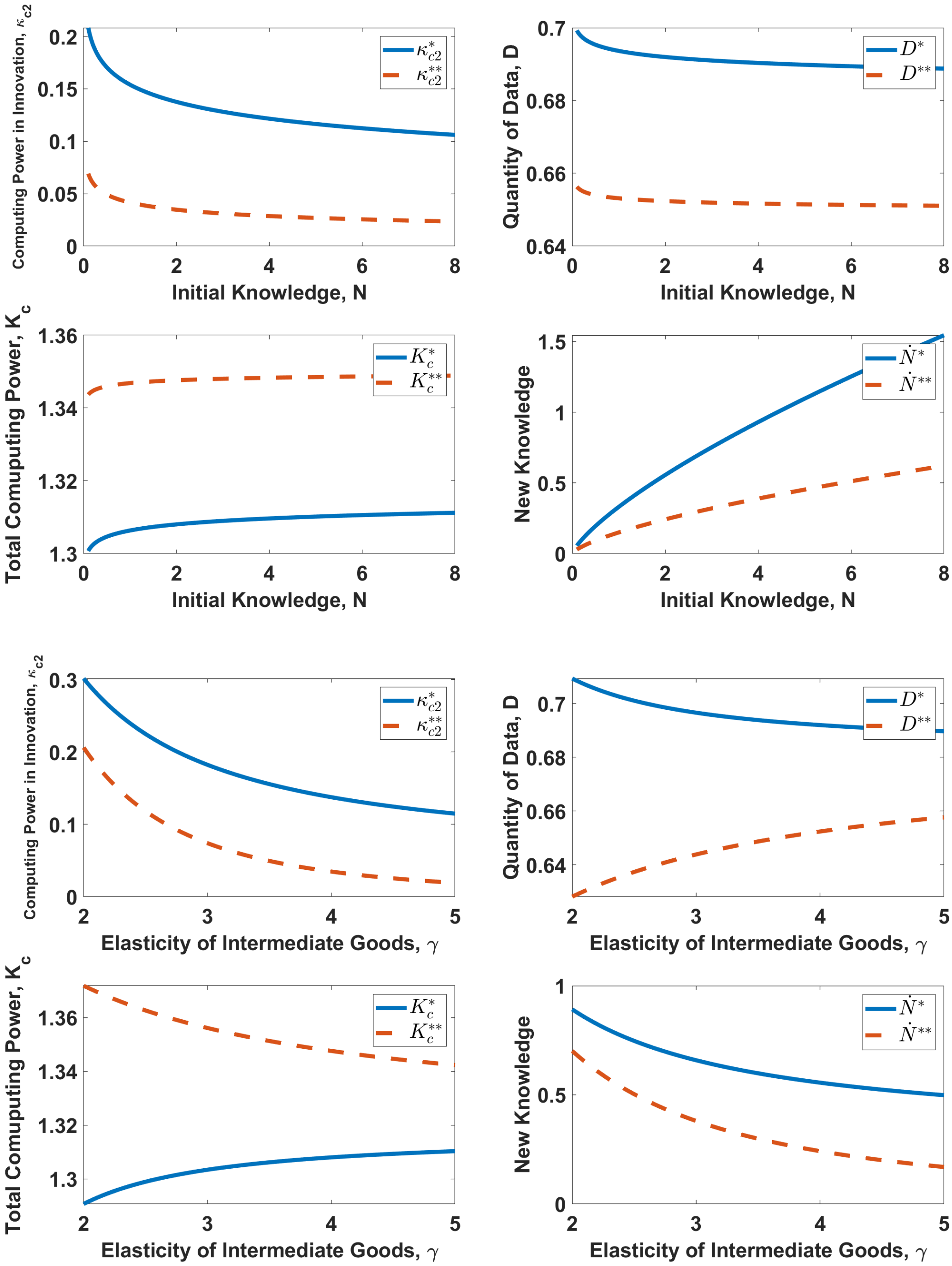


Figure 2: Comparative Statics with Respect to N_0 and γ

generative AIs compared to what's best for the society. This makes us rely more on data-driven the development of generative AI in the decentralized scenario, which leads to an excessive amount of data being used.

Overall, these insights show how small changes can have big impacts on how we balance our resources and foster the development of generative AI.

The simulation of ϕ provides valuable insights into the interplay between privacy costs, data sharing, and the development of generative AI in both equilibrium scenarios. Interestingly, our findings diverge from those of Cong et al. (2022), as our model introduces the role of computing power in shaping the outcomes of the digital economy.

In our analysis, we observe that higher privacy costs tend to discourage data sharing in both the social optimal and decentralized equilibria. This aligns with a common intuition that individuals are less willing to share data when privacy concerns increase. However, a notable distinction arises in how this effect interacts with the development of generative AI.

Unlike previous predictions from the literature that higher privacy costs lead to a decrease in both data amount and long-term economic growth, our model suggests a more nuanced outcome. The inclusion of computing power in the equation changes the dynamics. A digital economy, empowered by computing power, can counteract the negative impact of elevated privacy costs.

In our model, increased computing power enables more efficient utilization of available data, compensating for the reduced data sharing resulting from higher privacy costs. This efficient usage of data enhances the development of generative AI prospects, offsetting the potential stifling effect of increased privacy costs. As a result, the digital economy has the capacity to mitigate the adverse consequences predicted in settings where computing power is not considered.

This finding underscores the role of technological advancements, such as increased computing power, in reshaping the equilibrium outcomes in the face of changing privacy landscapes. It also underscores the importance of considering the broader technological context when analyzing the effects of privacy-related factors on the digital economy.

We also simulate the impact of enhanced storage technology growth on the digital economy, which is captured by the comparative statics of θ. In this context, as storage technology progresses, several noteworthy dynamics come into play, with significant implications for the overall equilibrium.

The growth of storage technology holds the promise of more precise and efficient knowledge recording, which, in turn, facilitates its broader dissemination and accelerated accumulation of the development of generative AI. This cascade of effects reverberates throughout the digital economy, leading to various outcomes as depicted in the figure.

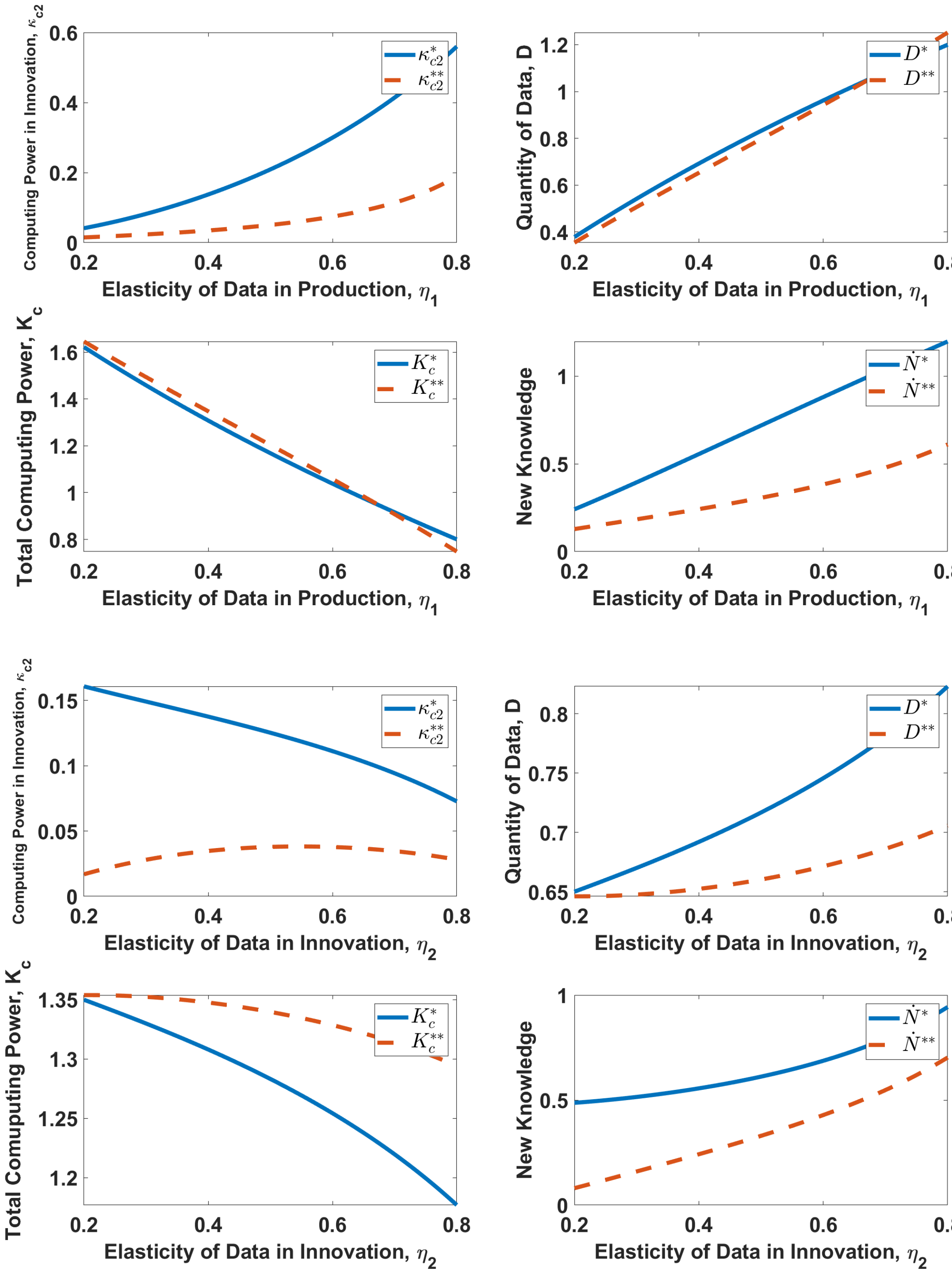


Figure 3: Comparative Statics with respect to the Output Elasticity of Data, η_1 and η_2

As storage technology improves, the efficiency of the the development of generative AI sector receives a boost. This heightened efficiency draws in more computing power, evident in the increase of κ_{c2}. Consequently, the the development of generative AI sector becomes more capable of offsetting the privacy costs associated with data sharing, thereby encouraging consumers to engage in more active data sharing. This outcome reflects the idea that when the benefits of sharing data, such as the development of generative AI, outweigh the concerns over privacy, consumers are more likely to participate.

Moreover, the increased data sharing has a multiplier effect on the productivity of the production sector due to the nonrivalrous nature of data. As more data becomes available and accessible, the production sector gains access to valuable insights, leading to productivity improvements and potentially even higher economic gains.

In essence, the comparative statics of θ underscores the relevance of technological advancements, data sharing, and productivity. It suggests that as storage technology continues to advance, the digital economy has the potential to experience a virtuous cycle of the development of generative AI, data sharing, and economic growth.

In an expanded scope of parameter variations encompassing χ_s, χ_c, η_1, and η_2, Figures 4(a) and 4(b) provide a comprehensive depiction of the disparities between the social optimal scenario and the decentralized equilibrium.

The observed trends in these figures indicate that the gap between the two cases tends to widen when the values of χ_s and χ_c are relatively modest, while the disparity narrows as these production costs increase. This observation is consequential, suggesting that when the costs associated with both storage and computing power production are high, the extent of market distortion is mitigated. Conversely, as production costs diminish, the welfare loss stemming from decentralized market dynamics becomes more pronounced. This insight underscores the relevance of closely monitoring scenarios where production costs are relatively low, as they are more susceptible to market inefficiencies that warrant regulatory scrutiny.

Furthermore, our model uniquely captures the potential for excessive data utilization, an aspect often overlooked in prior research that predominantly focuses on underutilization. This scenario is evidenced in the right subfigure of Figure 4(b), where a high value of η_1 coupled with a low value of η_2 manifests. In this configuration, an overallocation of computing power to the production sector leads to an excessive demand for data within the the development of generative AI sector. Such findings shed light on the complex interplay between parameter values, market dynamics, and their nuanced consequences for data utilization patterns.

In summary, Figures 4(a) and 4(b) provide a comprehensive understanding of the gap between the social optimal allocation and decentralized equilibrium. This analytical lens reveals the sensitivity of these dynamics to various parameter configurations and serves as a guide for policymakers to tailor interventions that address market inefficiencies, particularly in scenarios where production costs are relatively low.

4 Extended Discussion

In this section, we expand upon our baseline model, which primarily centers around computing power and data storage, to encompass algorithms as a distinct factor of production. To maintain clarity, we streamline the production process for data storage. The production of algorithms is contingent on both data and labor inputs and adheres to a Cobb-Douglas production function:

$$A = A_0^\delta D_a^\omega L_a^{1-\omega} \tag{26}$$

where δ denotes the weight assigned to prior experience in algorithm development, and ω characterizes the elasticity of substitution between data and labor inputs.

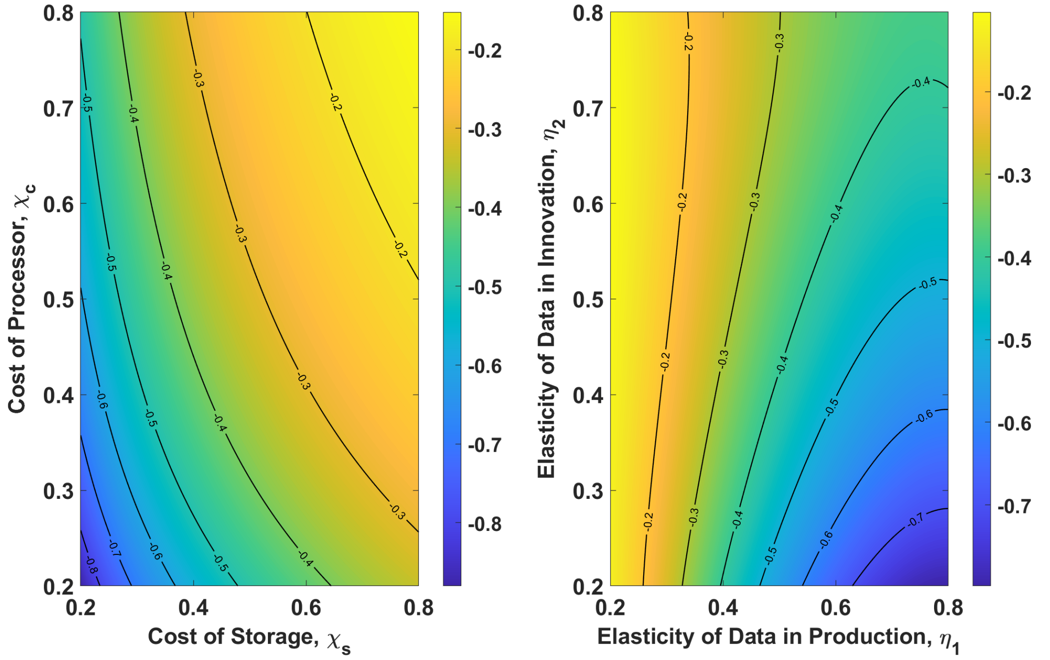
The central objective of the social planner remains the maximization of the aggregated utility across time periods, as expressed in the formulation:

$$\begin{aligned} & \max \ln c - \phi(d) \\ s.t. \qquad & c = Y = (N+\Delta N)^{\frac{\gamma}{\gamma-1}-(1-\eta_1)}(AD)^{\eta_1} K_{c1}^{1-\eta_1} \\ & A = A_0^\delta D_{3,t}^\omega L_a^{1-\omega} \\ & \Delta N = N_0^\theta (AD)^{\eta_2} K_{c2}^{1-\eta_2} \\ & \chi_c K_c + L_a = 1 \\ & K_{c1} + K_{c2} = K_c \end{aligned} \tag{27}$$

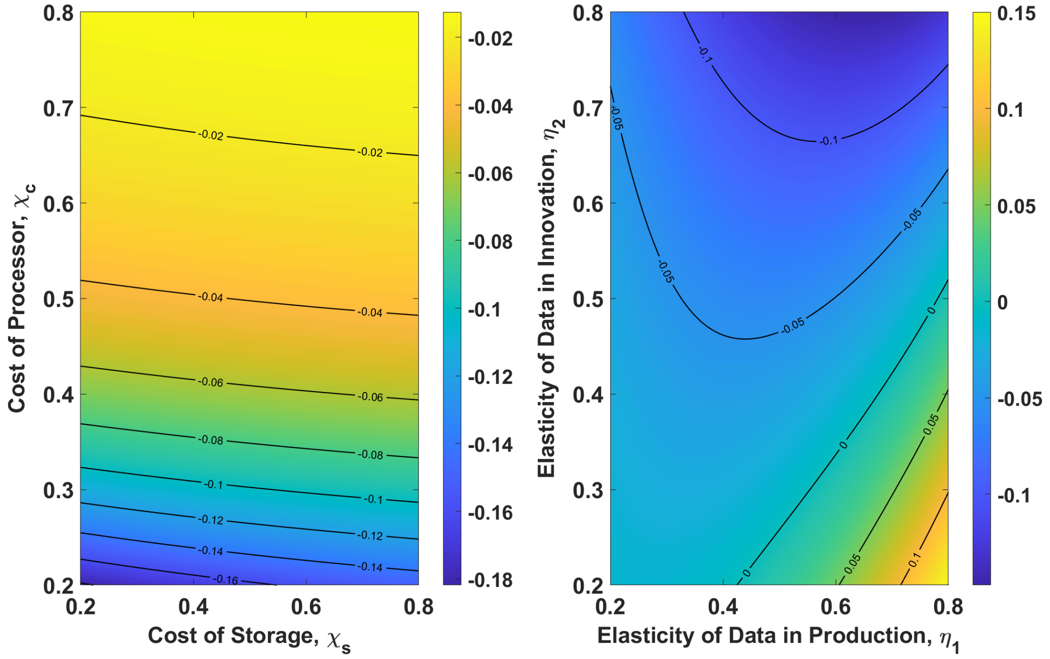
The optimal allocation $\{L_a^*, D^*, K_{c2}^*\}$ is determined by the following system of equations:

$$\begin{cases} \dfrac{\Gamma\omega\eta_2(N^*-N_0)}{N^* L_a^*} + \dfrac{(1-\omega)\eta_1}{L_a^*} = \dfrac{(1-\eta_2)(N^*-N_0)}{(1-L_a^*)} + \dfrac{1-\eta_1}{1-L_a^*} \\ \dfrac{\Gamma(1+\omega)\eta_2(N^*-N_0)}{N^*} + \dfrac{\omega\eta_1}{D^*} + \dfrac{1-\eta_1}{1-L_a^*} = \phi'(D^*) \\ \dfrac{\Gamma(1-\eta_2)(N^*-N_0)}{N^*\kappa_{c2}^*} = \dfrac{1-\eta_1}{1-\kappa_{c2}^*} \end{cases} \tag{28}$$

The set of first-order conditions outlined above encapsulates the allocation of labor and data inputs between algorithm development and computing power production, as well as the alloca-



(a) $N^{**} - N^{*}$



(b) $D^{**} - D^{*}$

Figure 4: Difference in the amount of knowledge and data under different χ_c, χ_s and η_2

tion of computing power between the the development of generative AI and production sectors. Notably, algorithms are non-rivalrous and can be harnessed by all sectors leveraging data, in contrast to computing power, which is characterized by rivalry.

Moreover, given the data-dependence of algorithm development, the return on data is heightened. In equilibrium, consumers are inclined to share more data, even as the privacy cost remains constant due to the recurring utilization of algorithms. This phenomenon aligns with the concept of “data laundry” articulated in Cong et al. (2021).

5 Conclusions

This paper introduces a comprehensive model encompassing data, computing power, storage, and algorithms to address a fundamental inquiry: how can labor, data, and computing power be allocated optimally in a digital economy? Our central proposition is that the incremental advantages of data sharing should align with its associated costs. The benefits of data sharing stem from its utility in both production and the development of generative AI domains, while its costs emanate from three main sources: the negative utility incurred due to privacy breaches for consumers, the opportunity cost associated with diverting labor from computing power production to data storage, impeding the development of generative AI, and the computing power dilution resulting from the expansion of data-sharing varieties.

Our analysis reveals that the implementation of monopolistic patents to incentivize the development of generative AI introduces distortions in data quantity, aggregated computing power, and factor allocation. The scarcity of the development of generative AI drives an excessive allocation of computing power to the production sector. Additionally, the output elasticity of data plays a significant role in shaping outcomes. In cases where data’s output elasticity is high in the production sector, optimal allocation channels more computing power toward production, reducing the the development of generative AI sector’s dependence on data through heightened data efficiency. In contrast, the decentralized economy allocates insufficient computing power to the development of generative AI and generates excessive data demand.

References

Abowd, John M and Ian M Schmutte (2019) “An economic analysis of privacy protection and statistical accuracy as social choices,” *American Economic Review*, 109 (1), 171–202.

Acemoglu, Daron, Ali Makhdoumi, Azarakhsh Malekian, and Asu Ozdaglar (2022) “Too much data: Prices and inefficiencies in data markets," *American Economic Journal: Microeconomics*, 14 (4), 218–256.

Aghion, Philippe, Benjamin F Jones, and Charles I Jones (2017) *Artificial intelligence and economic growth*, 23928: National Bureau of Economic Research Cambridge, MA.

Agrawal, Ajay, Joshua Gans, and Avi Goldfarb (2019) *The economics of artificial intelligence: an agenda*: University of Chicago Press.

Begenau, Juliane, Maryam Farboodi, and Laura Veldkamp (2018) "Big data in finance and the growth of large firms," *Journal of Monetary Economics*, 97, 71–87.

Cong, Lin William, Wenshi Wei, Danxia Xie, and Longtian Zhang (2022) "Endogenous growth under multiple uses of data," *Journal of Economic Dynamics and Control*, 141, 104395.

Cong, Lin William, Danxia Xie, and Longtian Zhang (2021) "Knowledge accumulation, privacy, and growth in a data economy," *Management Science*, 67 (10), 6480–6492.

Dierks, Ludwig and Sven Seuken (2022) "Cloud pricing: The spot market strikes back," *Management Science*, 68 (1), 105–122.

Fainmesser, Itay P, Andrea Galeotti, and Ruslan Momot (2023) "Digital privacy," *Management Science*, 69 (6), 3157–3173.

Farboodi, Maryam and Laura Veldkamp (2021) "A model of the data economy,"Technical report, National Bureau of Economic Research.

Goldfarb, Avi and Catherine E Tucker (2011) "Privacy regulation and online advertising," *Management science*, 57 (1), 57–71.

He, Yiyao, Shiyuan Pan, and Danyang Xie (2022) "Public Data Provision and Quality Upgrade in a Semi-Endogenous Growth Model with R&D," *Available at SSRN 4332937*.

Hou, Yao, Jinglei Huang, Danxia Xie, and Weidi Zhou (2022) "The Limits to Growth in the Data Economy: How Data Storage Constraint Threats," *Available at SSRN 4099544*.

Huang, Jinglei, Danxia Xie, and Yibai Yang (2023) "Data Right and Economic Growth," *Available at SSRN 4426537*.

Ichihashi, Shota (2021) "The economics of data externalities," *Journal of Economic Theory*, 196, 105316.

Jones, Charles I and Christopher Tonetti (2020) "Nonrivalry and the Economics of Data," *American Economic Review*, 110 (9), 2819–2858.

Marston, Sean, Zhi Li, Subhajyoti Bandyopadhyay, Juheng Zhang, and Anand Ghalsasi (2011) "Cloud computing—The business perspective," *Decision Support Systems*, 51 (1), 176–189.